3.4 - Concavity

Goals: Connect the idea of concavity to the second derivative and use this to find max and mins
Before Class Videos: 3.4.1

Definition:

A differentiable function $f(x)$ is

- concave up on an interval I if $f'(x)$ is increasing on I;
- concave down on an interval I if $f'(x)$ is decreasing on I.

Test for Concavity: If $f(x)$ is twice differentiable on an interval I,

- If $f''(x) > 0$ on I, then f is concave up on I;
- If $f''(x) < 0$ on I, then f is concave down on I.

Example 1: $y = x^3$:

Example 2:

$$f(x) = x^3 - 6x^2 + 15.$$

Theorem (2nd derivative test):

If f is continuous near c,

- if $f'(c) = 0$ and $f''(c) > 0$, f has a local min at c;
- if $f'(c) = 0$ and $f''(c) < 0$, f has a local max at c.

Definition:
A point $(c, f(c))$ on the graph of f is an inflection point if the concavity of f changes at c

Theorem:
If $(c, f(c))$ is an inflection point, then either $f''(c) = 0$ or $f''(c)$ does not exist

Example 3: $x^{1/3}$

Example 4: x^4

Example 5: Find all critical point and classify them for $3x^4 - 8x^3 - 90x^2$. Additionally find all inflection points.

Suggested after class problems from APEX: Section 3.4: 2,12,16